Benefit and risk of primary thromboprophylaxis in ambulatory patients with advanced pancreatic cancer receiving chemotherapy: a systematic review and meta-analysis of randomized controlled trials

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As vascular thromboembolism (VTE) is a leading cause of death in cancer patients, it has been postulated that primary thromboprophylaxis (PTP) in cancer patients might improve survival by reducing VTE occurrence. We performed a systemic review and meta-analysis of randomized controlled trials (RCTs) to investigate the benefit and risk of PTP with low-molecular-weight heparins (LMWHs) in ambulatory advanced pancreatic cancer (APC) patients receiving chemotherapy. We undertook a literature search using MEDLINE and EMBASE databases through May 2015. RCTs with reduction in symptomatic VTE as a primary or secondary endpoints were included. Mantel–Haenszel method was used to estimate the pooled event-based risk ratio as well as the pooled absolute risk difference with 95% confidence interval (CI). Seven hundred and thirty-eight APC patients were eligible for analysis. PTP lasted 3–6 months. The crude VTE incidence was 2.1 and 11.2% in LMWH and in control groups, respectively (risk ratio, 0.18; 95% CI, 0.083–0.39; $P<0.0001$). The absolute risk difference in VTE was −0.092 (95% CI, −0.127 to −0.057; $P<0.0001$), with an estimated number needed to treat of 11 patients to prevent one symptomatic VTE event. The pooled risk ratio for major bleeding was 1.25 (95% CI, 0.48–3.3, $P=0.65$). Although these findings are encouraging to deploy PTP in APC patients receiving chemotherapy, uncertainties remain as to its survival benefit, optimal PTP duration, type and dose of LMWH, and costs of care. Therefore, adequately powered randomized phase III studies are warranted to address these questions. *Blood Coagul Fibrinolysis* 27:270–274 Copyright © 2016 Wolters Kluwer Health, Inc. All rights reserved.

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Introduction

Patients with cancer are at a significantly increased risk of developing venous thromboembolism as well as arterial thromboembolic events [1–3]. It is estimated that approximately 20–30% of all first venous thromboembolic events are associated with cancer [3]. Vascular thromboembolism (VTE) imposes significant morbidity and mortality in cancer patients. It is the second leading cause of death in patients with cancer after cancer progression itself [4]. The 1-year survival rate of cancer patients with VTE was one-third that of cancer patients without VTE in one study [5]. In addition, cancer patients have a higher incidence of VTE recurrence and bleeding complications during anticoagulant therapy than patients without cancer [6]. As mortality is increased in cancer patients who develop VTE compared with those without VTE [2,5,7], it has been postulated that primary thromboprophylaxis (PTP) in cancer patients might improve survival and quality of life by reducing VTE occurrence. However, routine prophylaxis against VTE in ambulatory patients with solid cancer remains extremely controversial. Risk factors for cancer-associated thrombosis could be divided into patient-specific, cancer-specific and treatment-specific risk factors [8]. To name a few, these include certain types of cancer such as pancreatic and haematologic malignancies, advanced-stage cancer and antineoplastic therapy such as chemotherapy and radiotherapy [8,9]. The more risk factors a cancer patient has, the more likely the patient is to experience VTE [10]. It is therefore anticipated that such a high-risk patient has a more favourable benefit/risk ratio with PTP. Patients with advanced pancreatic cancer receiving chemotherapy fall into this high-risk category, and thus, to investigate the benefit and risk of PTP in such patients, we carried out a systematic review and meta-analysis of randomized controlled trials (RCTs).

Materials and methods

Literature search

We undertook a systematic literature search using MEDLINE and EMBASE databases up to 20 May 2015. Both databases were accessed through www.embase.com using the keywords 'thromboprophylaxis OR anticoagulation OR warfarin OR heparin AND ('cancer'/exp OR cancer)' using Embase mapping options (Map to preferred term in Emtree; Search also as free text in all fields; Explode

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using narrower Emtree terms; and Search as broadly as possible). The search was limited to 'controlled clinical trials' and 'humans'. Studies written in both English and non-English languages were searched. The titles and abstracts of the resulting citations were reviewed. Full-text articles of potential studies were retrieved to assess eligibility. References of the potential studies and clinical guidelines on thromboprophylaxis in cancer patients were also reviewed for any additional relevant studies.

Eligibility criteria

Studies that were eligible for inclusion in the meta-analysis had to fulfil all of the following criteria: RCTs of patients who were receiving chemotherapy for locally advanced or metastatic pancreatic cancer; RCTs comparing anticoagulation therapy and a control group; and symptomatic VTE events (sVTE) during the anticoagulation therapy as a primary or secondary efficacy endpoint of the RCT.

Data extraction and study outcome measures

Two authors (N.M.T. and E.G.) independently conducted data extraction from each eligible study. The following data were collected: first author's last name; year of publication; acronym of the study if available; study type; its primary and secondary endpoints; types of malignancy enrolled in the study; study drug; its dosage and duration; number of VTE events, death and major bleeding episodes in the experimental and control arms; number of patients included in each arm; and overall survival data. Any discrepancies were addressed by a joint re-evaluation with the third author. The primary outcome of our meta-analysis was sVTE. The secondary outcome was major bleeding events.

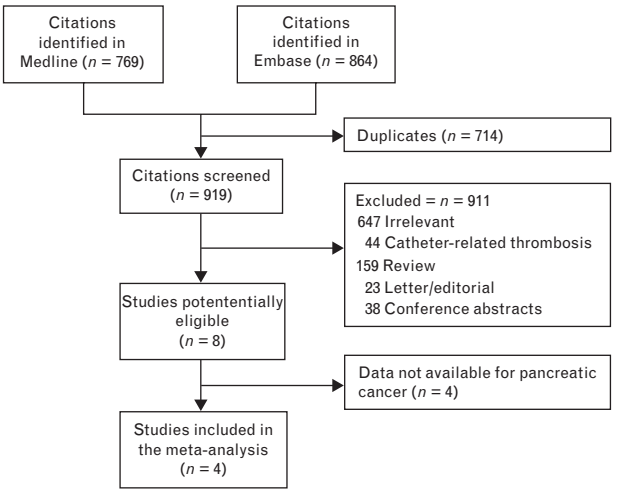
Quality assessment

The quality of the included studies was independently evaluated by two authors (N.M.T. and E.G.) using the Newcastle–Ottawa Scale (NOS) [11]. The NOS assigns a maximum of four points for selection, two points for comparability and three points for exposure/outcome, and thus nine points on the NOS indicate the highest study quality.

Data synthesis and analysis

Mantel–Haenszel method was used to estimate the pooled event based risk ratio as well as the pooled absolute risk difference with 95% confidence interval (CI). Fixed effects model was applied because there was no significant heterogeneity among the included studies. All statistical analyses and graphs were obtained by using Comprehensive Meta Analysis Version 2.2.064 (Biostat, Englewood, New Jersey, USA). The level of significance was set at P less than 0.05. Data from this meta-analysis are presented in accordance with the checklist proposed by the PRISMA Statement [12].

Fig. 1



Flow diagram of the study selection process.

Results

Search results

Two RCTs [13,14] and a subgroup of another two RCTs [15,16], comprising a total of 738 patients with advanced pancreatic cancer, were eligible for inclusion in our meta-analysis. Figure 1 illustrates the detailed steps of study selection.

Characteristics of the studies

The characteristics of the included studies are depicted in Table 1. Anticoagulants used in these trials were nadroparin (prophylactic dose), semuloparin (prophylactic dose), enoxaparin (semi-therapeutic dose) and dalteparin (therapeutic dose). PTP was given for the duration of chemotherapy, that is mostly around 3 months.

Meta-analysis results

The crude incidence of sVTE was 2.1 and 11.2% in patients receiving PTP and in the control group, respectively, with a resultant risk ratio of 0.18 (95% CI, 0.083–0.39; $P < 0.001$) (Fig. 2). The absolute risk difference in sVTE was −0.092 (95% CI, −0.127 to −0.057; $P < 0.001$) (Fig. 3), with an estimate of the number needed to treat (NNT) of 11 patients (95% CI, 8–18; $P < 0.001$) to prevent one sVTE. Major bleeding events were reported in 4.1% of patients on PTP compared with 3.3% in control patients according to an analysis of two RCTs [13,14]. The pooled relative risk for major bleeding was statistically nonsignificant at 1.25 (95% CI, 0.48–3.3, $P = 0.65$) (Fig. 4).

Quality assessment results

The quality assessment results of the individual studies are summarized in Table 1. The average NOS score was 7.3 out of 9.

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Table 1 Characteristics of the trials included in the meta-analysis

Study	Total number of patients with pancreatic cancer	Intervention	Primary efficacy outcome measure	Primary safety outcome measure	Time of initiation of thromboprophylaxis	Type of chemotherapy	Study Type	NOS Score
SAVE-ONCO (Agnelli *et al.* [15], subgroup)	254	Semuloparin 20 mg SC daily x duration of chemotherapy	sVTE during the study drug treatment and 3 days	Clinically relevant bleeding that occurred up to 3 days after the last dose of the study drug	At the start of chemotherapy	Data not available	Randomized, placebo-controlled, double-blind, multicenter study	8
FRAGEM (Maraveyas *et al.* [13])	121[a]	Dalteparin 200 IU/kg SC daily x 4 weeks, then 150 IU/kg x 8 weeks	sVTE in <100 days of treatment initiation	None planned	At the start of chemotherapy	Gemcitabine	Randomized controlled phase 2b trial	7
PROTECHT (Agnelli *et al.* [16], subgroup)	53	Nadroparin 3800 IU SC daily x duration of chemotherapy or maximum 120 ± 10 days	sVTE during the study drug treatment and 10 days	Major bleeding that occurred up to 48 h after the last dose of the study drug	At the start of chemotherapy	Data not available	Randomized, placebo-controlled, double-blind, multicenter study	7
CONKO 004 (Pelzer *et al.* [14])	312	Enoxaparin 1 mg/kg SC daily x 3 months, then 40 mg daily for 3 months	sVTE in the first 12 weeks of treatment	None planned	At the start of chemotherapy	Combination of gemcitabine, 5-fluorouracil, folinic acid and cisplatin (GFFC) or gemcitabine monotherapy	Prospective, open-label, randomized, multicenter and group-sequential phase 2b trial	7

NOS, the Newcastle–Ottawa Scale; SC, subcutaneously; sVTE, symptomatic venous thromboembolism. [a] 119 patients were evaluable for incidence of sVTE.

Heterogeneity and publication bias

There was mild between-study heterogeneity for risk ratio and risk difference analysis for VTE events ($I^2 = 34$ and 38%, respectively). No heterogeneity was observed for risk ratio analysis for major bleeding events ($I^2 = 0\%$). There was no evidence of publication bias detected for risk ratio or risk difference analysis by either the Begg and Mazumdar rank correlation test or the Egger's test of the intercept.

Discussion

The results of our meta-analysis indicate that thromboprophylaxis in advanced pancreatic cancer patients receiving chemotherapy resulted in an 82% reduction in the relative risk of sVTE with NNT of 11. This is in contrast to the finding of a previous meta-analysis that reported approximately 60 patients were required to be treated with low-molecular-weight heparin (LMWH) to prevent one sVTE among unselected cancer patients receiving chemotherapy [17]. The discrepant finding in the benefit of PTP suggests that high-risk patient selection is of crucial importance to improve outcome. In addition, the observed risk reduction in our meta-analysis was not associated with an increase in major bleeding, implying that PTP with LMWH may be beneficial in reducing VTE events in ambulatory patients with advanced pancreatic cancer undergoing chemotherapy. We were unable to calculate the event rates as a function of time on study, as the published PROTECHT and SAVE-ONCO trials did not describe the VTE events in details [15,16]. Interestingly, symptomatic VTE event rates on LMWH group continue to decrease compared with control group even after the discontinuation of LMWH in the FRAGEM (9% dalteparin arm vs. 22% control arm at >100 days) and CONKO-004 (5.7% enoxaparin arm vs. 15.1% control arm at 12 months) trials [13,14].

In a real-world analysis of venous thromboembolism risk in patients with cancer receiving chemotherapy, 11.6% (221 out of 1912) of patients with pancreatic cancer developed venous thromboembolism within 3.5 months after starting chemotherapy [18]. Our meta-analysis revealed a comparable pooled incidence rate of sVTE (11.2%, 40 out of 357) in the control group around the same time frame. The similarity in these two patient populations reflect that the efficacy of LMWH in preventing sVTE as evident in our meta-analysis might be applicable to the real-world pancreatic cancer patients.

It is noted that a previous meta-analysis of PTP in solid cancer patients had included a subset analysis of pancreatic cancer patients [19]. However, there are several concerns regarding the consistency and accuracy of the data collected in that analysis such as inclusion of arterial thromboembolic events despite the meta-analysis's primary endpoint of venous thromboembolism; inclusion of asymptomatic as well as symptomatic VTE events;

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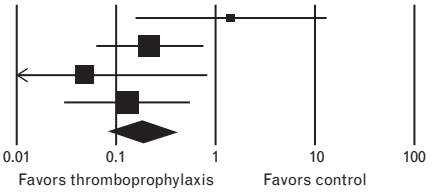
Fig. 2

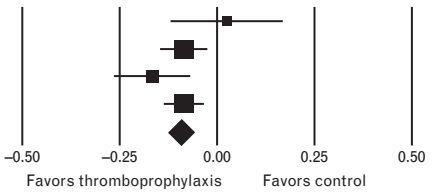
Study name	Statistics for each study					sVTE events / total	
	MH risk ratio	Lower limit	Upper limit	*Z*-value	*P*-value	Thromboprophylaxis	Control
Agnelli 2009 PROTECHT	1.417	0.159	12.638	0.312	0.755	3 / 36	1 / 17
Agnelli 2012 SAVE-ONCO	0.218	0.064	0.739	–2.445	0.015	3 /126	14 /128
Maraveyas 2012 FRAGEM	0.048	0.003	0.808	–2.109	0.035	0 / 59	10 / 60
Pelzer 2015 CONKO-004	0.127	0.029	0.545	–2.776	0.005	2 / 160	15 / 152
	0.180	0.083	0.390	–4.354	0.000	8 / 381	40 / 357



Pooled risk ratio for symptomatic vascular thromboembolism in patients with advanced pancreatic cancer receiving thromboprophylaxis vs. control.

Fig. 3

Study name	Statistics for each study					sVTE events / total	
	MH risk difference	Lower limit	Upper limit	*Z*-value	*P*-value	Thromboprophylaxis	Control
Agnelli 2009 PROTECHT	0.025	–0.119	–0.168	0.334	0.738	3 / 36	1 / 17
Agnelli 2012 SAVE-ONCO	–0.086	–0.146	–0.025	–2.783	0.005	3 /126	14 /128
Maraveyas 2012 FRAGEM	–0.167	–0.264	–0.069	–3.351	0.001	0 / 59	10 / 60
Pelzer 2015 CONKO-004	–0.086	–0.137	–0.036	–3.349	0.001	2 / 160	15 / 152
	–0.092	–0.127	–0.057	–5.116	0.000	8 / 381	40 / 357



Pooled risk difference for symptomatic vascular thromboembolism in patients with advanced pancreatic cancer receiving thromboprophylaxis vs. control.

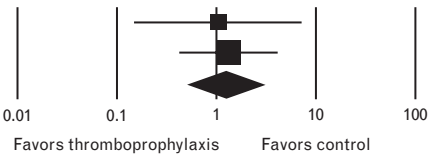
VTE events on long-term follow-up beyond the treatment period with anticoagulant; inaccurate number of patients in the control group used for VTE incidence analysis for the FRAGEM trial; and inaccurate number of VTE events for the PROTECHT trial [19]. In contrast, our meta-analysis strictly followed the eligibility criteria and included only the symptomatic vascular thromboembolic events that occurred around the time of PTP in advanced pancreatic cancer patients receiving chemotherapy (Table 1).

Although there is a significant benefit conferred by PTP in terms of reduction in VTE events without increasing major bleeding episodes, it seems that there is no survival benefit of such intervention. In the FRAGEM trial, 100-day mortality was 11% in the control arm and 7% in the PTP arm ($P = 0.39$) [13]. Similarly, CONKO-004 trial reported no difference in median overall survival between the control and PTP arms (8.02 vs. 8.51 months, respectively, $P = 0.44$) [14]. Sufficient data were not available from the included RCTs in order to perform a survival analysis.

In addition, because pancreatic adenocarcinoma is a very aggressive type of cancer with an extremely high case fatality rate and a median overall survival of 4 months [20], the benefit of reduced VTE occurrence might not confer an improvement in overall survival in the same way as it was seen with cancer patients in general [5].

Fig. 4

Study name	Statistics for each studay						
	MH risk ratio	Lower limit	Upper limit	*Z*-value	*P*-value	Thromboprophylaxis	Control
Maraveyas 2012 FRAGEM	1.051	0.153	7.220	0.050	0.960	2 / 59	2 / 62
Pelzer 20154 CONKO-004	1.330	0.431	4.101	0.496	0.620	7 / 160	5 / 152
	1.253	0.475	3.309	0.455	0.649	9 / 219	7 / 214



Pooled risk ratio for major bleeding in patients with advanced pancreatic cancer receiving thromboprophylaxis vs. control.

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It is at present unclear whether extended PTP beyond the period of chemotherapy is of benefit in patients with advanced pancreatic cancer. In CONKO-004 trial, it was found that the beneficial effect of short-term PTP (semi-therapeutic dose of enoxaparin for 3 months) on sVTE decreased by three-fold over time (relative risk of 0.13 at 3 months vs. 0.38 at 12 months) [14]. Likewise, FRAGEM trial showed that beyond the period of anticoagulation, the relative risk for sVTE increased by eight-fold over time (relative risk of 0.048 at 100 days vs. 0.39 at 12 months) [13]. These findings suggest that extended PTP might be useful in reducing sVTE events at 12 months.

Our study has several limitations that should be acknowledged. The studies included used different anticoagulants as well as differing dose regimens (Table 1); it is unclear regarding the equivalency of the efficacy and safety of therapeutic dose vs. those of semi-therapeutic or prophylactic dose of anticoagulants with regard to sVTE prevention.

In summary, our results indicate that targeted PTP resulted in an 82% reduction in the relative risk of sVTE without increasing major bleeding events, and that 11 patients were needed to be treated to prevent one sVTE during chemotherapy. Although these findings are encouraging to deploy PTP in patients with advanced pancreatic cancer receiving chemotherapy, uncertainties remain as to its survival benefit, optimal duration of PTP, type and dose of LMWH, quality of life and costs of care. Therefore, adequately powered randomized phase III studies are warranted to address these questions.

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N.M.T. and E.G. did the conception and design, data collection and analysis, and writing of the manuscript. T.H.O. did the conception and critical comments on the manuscript. All authors read and approved the final manuscript.

IRB approval was not applicable for this study (This meta-analysis study was presented as an abstract at the annual meeting of the American Society of Hematology in December 2014).

Conflicts of interest

There are no conflicts of interest.

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